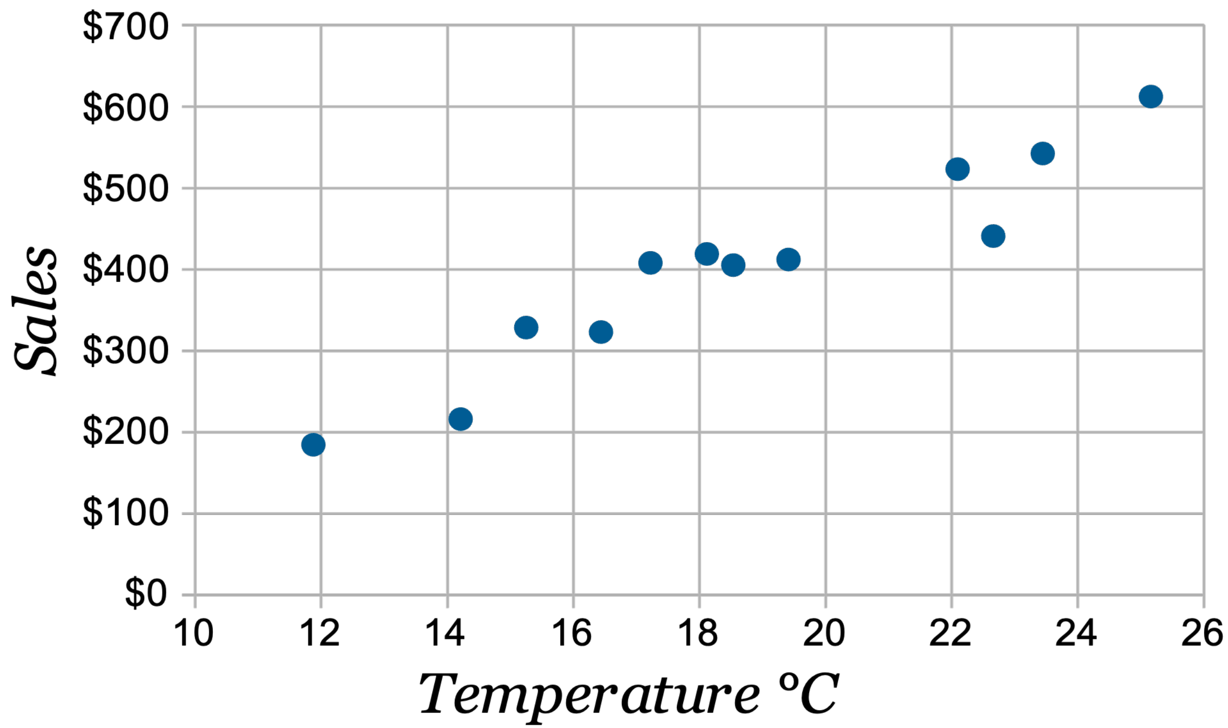
Ch 4. Bivariate Data

- Introduction to Bivariate Data
- Pearson Correlation and Covariance
- Properties of Person Correlation
- Variance Sum Law II

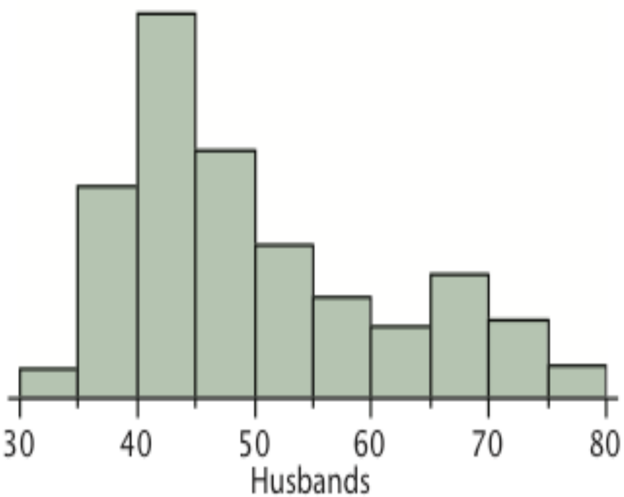
- Bivariate Data

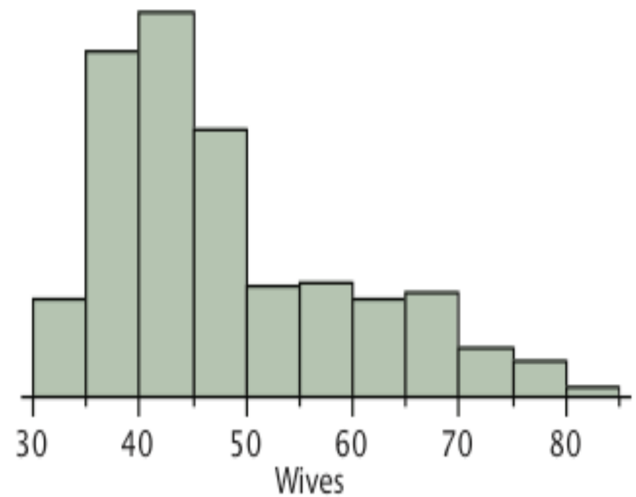
  - A dataset with a pair of variables which may be correlated to one another.

Eg: two variables – ice cream sales and temperature



- Bivariate Data



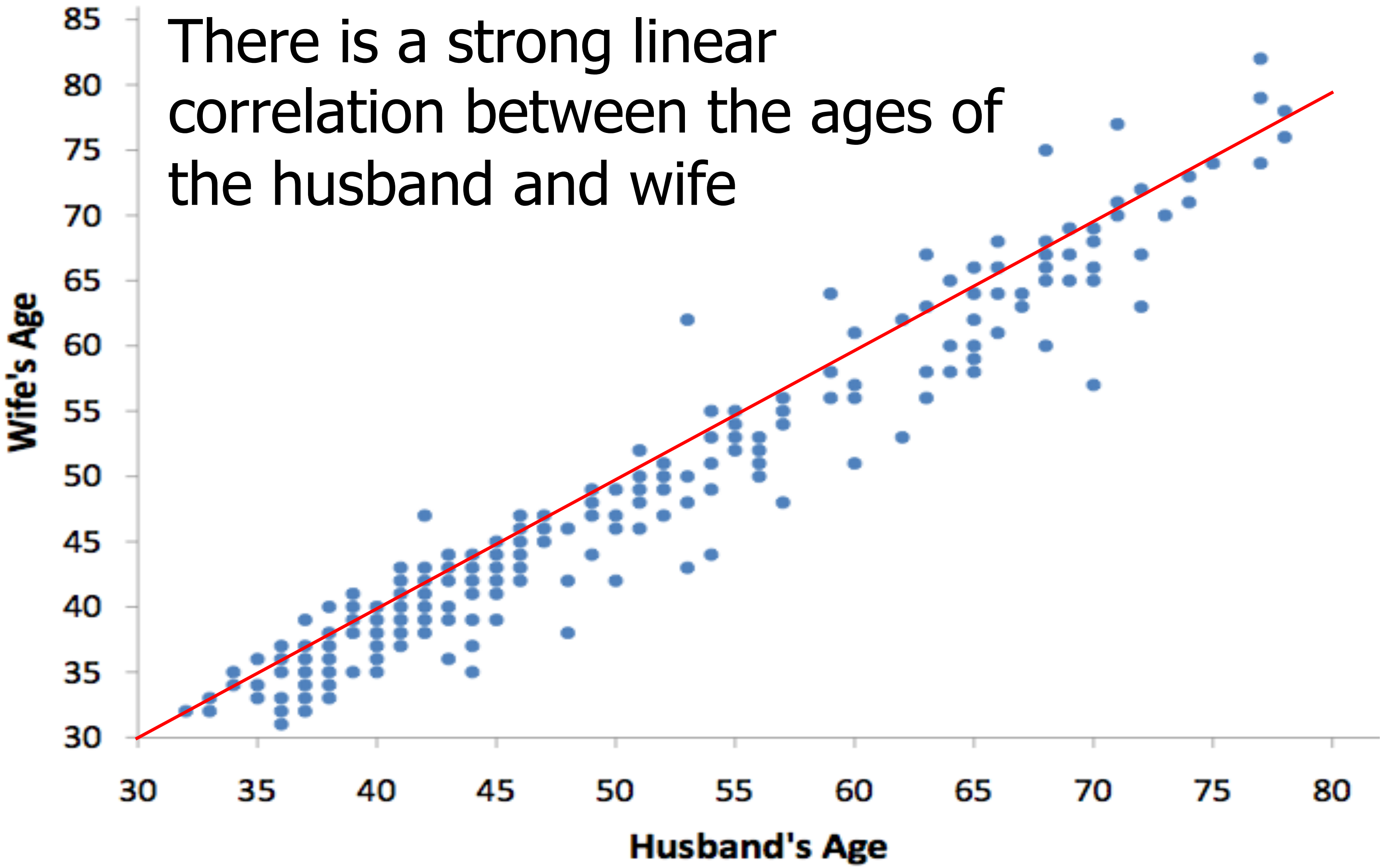


Indication on the central tendency, variability and the shape of distribution of ages.

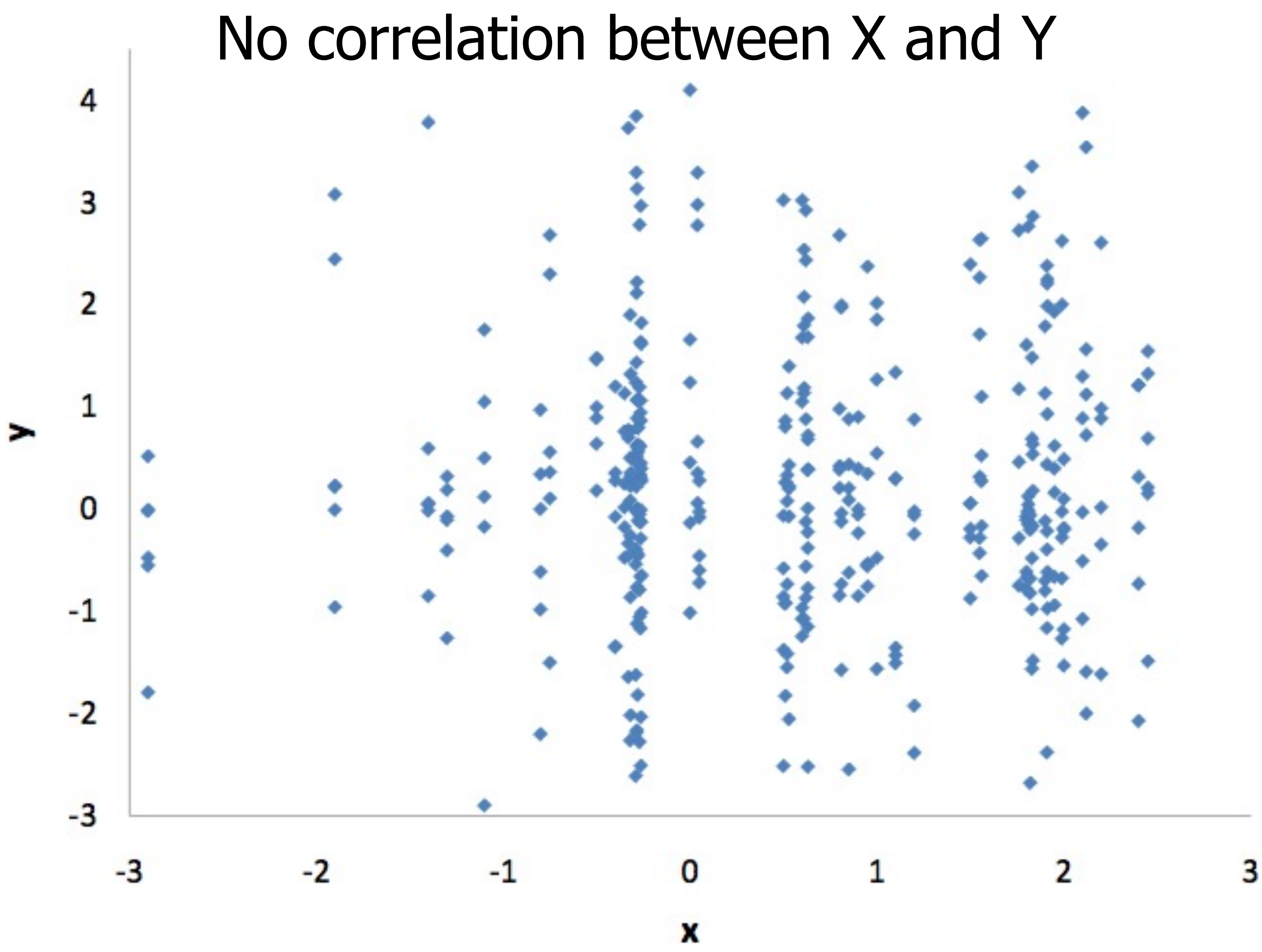
No indication on the correlation of the 2 variables.

- Bivariate Data

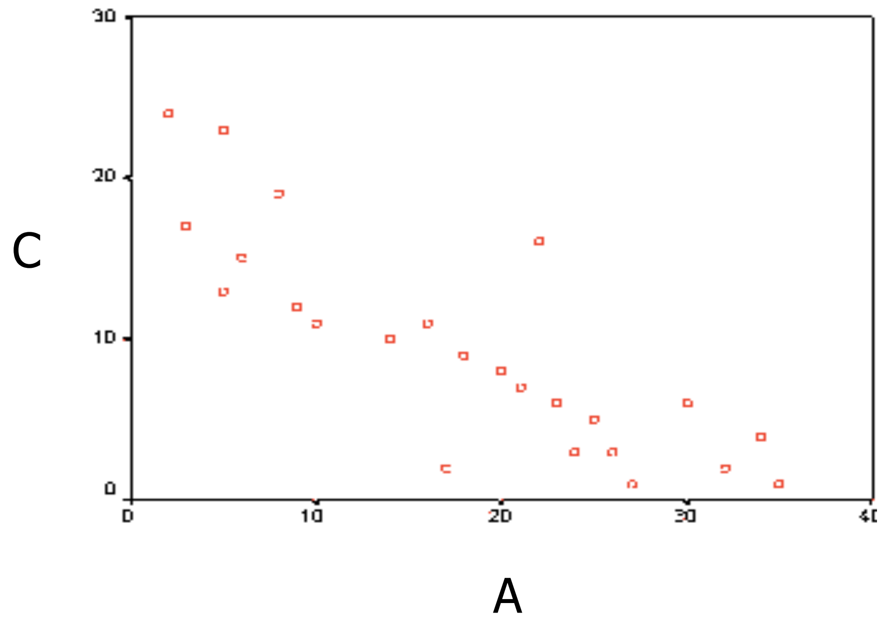
There is a strong linear correlation between the ages of the husband and wife



Eg: Correlation of variables X and Y



Describe the relationship between variables A and C. What could these variables represent in real life?



Negative relationship between A and C.
There is a negative relationship between price and quantity of the products that we buy.

- Pearson Correlation ρ

- An indicator on the strength of the <u>linear</u> relationship between two variables.

Definition: $\rho = \dfrac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$

Covariance of X and Y, denoted as cov(X,Y)

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$$= \frac{E[XY]-\mu_X\mu_Y}{\sqrt{E[X^2]-(\mu_X)^2}\sqrt{E[Y^2]-(\mu_Y)^2}}$$

$$= \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}}\sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}}$$

If $\mu_X = \mu_Y = 0$, then $\rho = \dfrac{\sum XY}{\sqrt{\sum X^2}\sqrt{\sum Y^2}}$

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Eg: Given the data set, calculate σ_X, σ_Y, cov(X,Y) and the Pearson Correlation.

X	2	5	6	8	9
Y	8	5	2	4	1

Given the data shown in the figure, is it appropriate to use Pearson Correlation to describe the relationship between X and Y?

- Computation of Correlation based on a sample of size n

$$\text{cov(XY)} = \frac{1}{n-1}\sum(X-\bar{X})(Y-\bar{Y})$$

Correlation $r = \dfrac{E[(X-\bar{X})(Y-\bar{Y})]}{s_X s_Y}$

$$= \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{n}}\sqrt{\sum Y^2 - \frac{(\sum Y)^2}{n}}}$$

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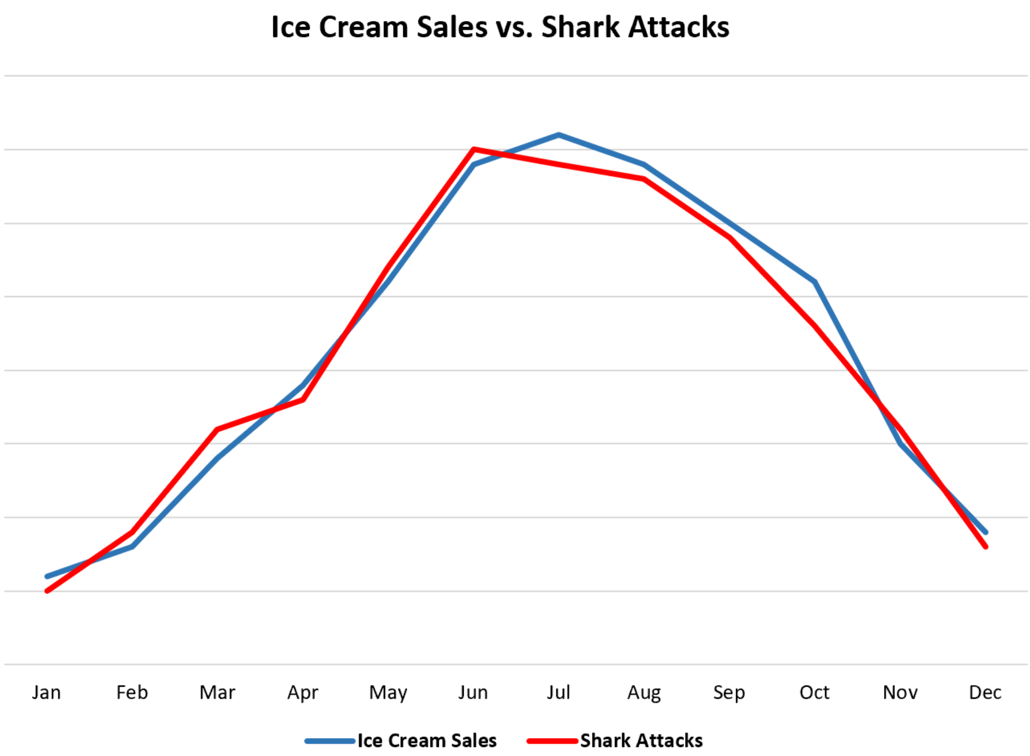
Correlation is not causation.



- Properties of Correlation

- Value in the range of [–1, +1]

- Symmetric: correlation of X with Y
= correlation of Y with X

- Unaffected by linear transformations:

Correlation of Y with X
= correlation of Y with A X + B
where A and B are constants

Eg: If the correlation between weight (in pounds) and height (in feet) is 0.58, find:

(a) the correlation between weight (in pounds) and height (in yards)

(b) the correlation between weight (in kilograms) and height (in meters).

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The correlation for both (a) and (b) is still 0.58 because linear transformations do not affect the value of Pearson's correlation, and both of the above instances are linear transformations.

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    E.g. $\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 + 2\overset{+}{\rho}\sigma_x\sigma_y > \sigma_x^2 + \sigma_y^2$

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- For computation based on a sample

$$s_{x\pm y}^2 = s_x^2 + s_y^2 \ \pm 2r s_x s_y$$

Eg: Students took 2 parts of a test, each worth 50 points. Part A has a variance of 25, and Part B has a variance of 49. The correlation between the test scores is 0.6.

(i) If the teacher adds the grades of the two parts together to form a final test grade, what would the variance of the final test grades be?

(ii) What would the variance of Part A - Part B be?

(i) $\text{Var}(A + B) = 25 + 49 + 2 * 0.6 * \sqrt{25} * \sqrt{49}$

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$= 116$

(ii) $\text{Var}(A - B) = 25 + 49 - 2*0.6*\sqrt{25}*\sqrt{49}$
$= 32$